

healthiar: An R package to quantify and monetize health impacts attributable to (environmental) exposure

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Summary

Health impacts attributable to exposures, such as ambient air pollution or noise, can be quantified (and monetized). However, such assessments can be time-consuming in terms of programming and often lack transparency in methodology. The open-source R package healthiar contributes to solving these challenges by providing functions to quantify (and monetize) attributable health impacts. It offers multiple calculation pathways and options, including iteration across multiple geographical units and stratification by subgroups. Epidemiologists and public health professionals can benefit from using healthiar for health impact quantifications.

Statement of need

Environmental exposures, e.g., ambient air pollution, are major contributors to disease and mortality worldwide ([GBD 2023 Disease and Injury and Risk Factor Collaborators, 2025](#)). Epidemiologists and public health researchers or practitioners quantify the (negative or positive) health impacts that are attributable to (no) changes in exposure to inform policy making.

The general methodology for quantifying health impacts has been documented in the literature ([Lehtomäki et al., 2025](#)). However, studies of specific assessments are not always transparent or complete about the specific methods used and the assumptions made, hindering reproducibility and plausibility checks. Additionally, the calculation can become tedious and error-prone, also when programming, e.g., in R ([R Core Team, 2023](#)).

The open-source R package healthiar quantifies (and monetizes) health impacts attributable to exposure. healthiar contributes to the comparability and transparency of results by using established methodology approved by international experts and making it publicly available. The package is available on the Comprehensive R Archive Network (CRAN), with a development version accessible via a public [GitHub repository](#). Online documentation is provided on the [package website](#).

healthiar enables multiple calculation pathways and options. Health impacts can be quantified using a) relative or absolute risk, b) single or sex- and age-specific baseline health data, c)

fixed-shape (e.g. linear, log-linear...) or user-defined exposure-response functions, d) population-weighted mean or categorical exposure. Additionally, `healthiar` offers the option to specify disability weight, cutoff and confidence interval, stratify by socio-economic group, iterate assessments across geographical units, compare scenarios, summarize uncertainty using Monte Carlo simulation, monetize health impacts or consider social inequalities in exposures. While `healthiar` focuses on ambient air pollution and noise exposure, it can potentially be used for other environmental or non-environmental exposures (e.g., green spaces, chemicals or physical activity).

State of the field

Existing R packages and tools for quantifying health impacts do not fully meet the needs of `healthiar` due to fundamental differences in scope and design:

- The World Health Organization (WHO) tool AirQ+ (Amini et al., 2024) focuses on ambient air pollution, but is offline and not an R package, limiting integration with programming workflows. While it partly overlaps with `healthiar` (and used to validate some `healthiar` results), it lacks the applicability for other exposures.
- The R package `rhap` (Sampedro et al., 2025) targets household air pollution using predefined scenarios from the Global Change Analysis Model, lacking the applicability for ambient exposures and custom scenarios offered by `healthiar`.
- The R package `ithimr` (Abbas et al., 2023), based on ITHIM (Integrated Transport and Health Impact Modelling tool), and the online WHO tool HEAT (Health-Economic Assessment Tool for walking and cycling) (Götschi et al., 2020) use walking and cycling data to quantify health impacts (from physical activity, air pollution, and traffic crashes). Instead, `healthiar` derives health impacts directly from environmental exposures, enabling broader applications beyond transport. Additionally, HEAT (like AirQ+) lacks an R package counterpart, limiting integration in programming workflows.

Extending these R packages and tools was impractical because it would have required major architectural changes, distorting their original purpose and complicating usage. Thus, `healthiar` was developed as a standalone R package to provide flexibility, modularity, and programming integration.

Software design

The functions included in `healthiar` enable a modular workflow (Figure 1). Functions for additional analyses, such as modification of existing scenarios, scenario comparisons, or multi-exposure assessments, use `healthiar` outputs as input data. In other cases, for instance, for monetization and social inequalities, the health impact can be entered either using `healthiar` outputs or directly from user input. Furthermore, smaller helper functions covering preparatory steps or single calculation steps are available.

healthiar : : CHEAT SHEET

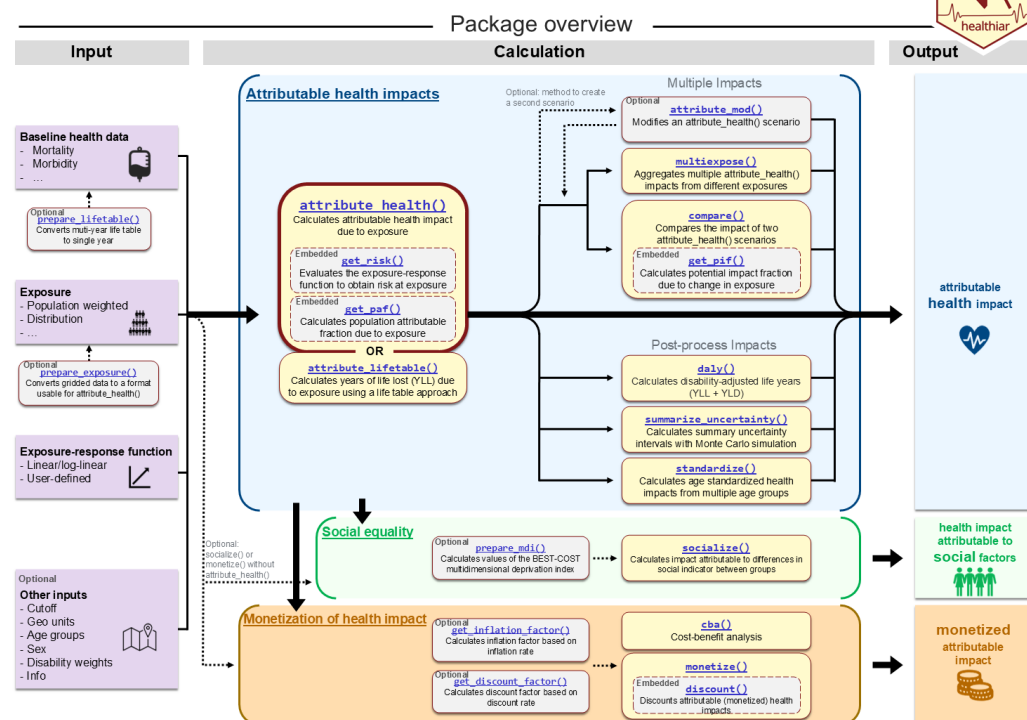


Figure 1: Overview of the healthiar functions (June 2026 snapshot; latest version on the healthiar website).

We refrained from providing default or background input data (e.g., exposure data, exposure-response functions, or baseline health data), as they a) would be incomplete given the many different exposures and health outcomes and b) would become quickly outdated, and updates would be time-consuming. Nevertheless, the R package includes example datasets that allow users to explore the functions.

Input data for healthiar functions are entered as separate arguments using vectors instead of within a data set/frame. This has two objectives: a) to facilitate simple assessment without creating a table, and b) to explicitly identify input data, avoiding mismatches.

The use of tidy data (Wickham, 2014) is supported and encouraged by healthiar. Thus, each row in the user data set corresponds to a unique stratification level (geographical unit, social group...), and each column to a function argument.

Methodological decisions have been taken together with experts from the European project BEST-COST (European Commission, 2025a). Currently, more than 350 internal tests check the correct behavior of healthiar, including the correctness of its results.

Research Impact Statement

Still as a non-public prototype, healthiar was firstly presented at the scientific conference Urban Transitions in November 2024 (Castro, 2024). In September-October 2025, shortly after the public release on GitHub, two international online workshops introduced healthiar to around 110 participants. healthiar was published in CRAN on 11 November 2025 and, eight months later, had been downloaded over 2,000 times (Csárdi, 2026).

Within the European research project BEST-COST (where healthiar was developed), the

package was adopted to quantify health impacts attributable to air pollution and noise in case studies (publications in preparation). External adoption of healthiar (beyond BEST-COST) further demonstrates its significance as a research tool:

- The European research project UBD-Policy (European Commission, 2025b) also adopted healthiar for case studies on air pollution, e.g. in Barcelona (Cussotto et al., 2025).
- Other researchers deployed healthiar to quantify health impacts attributable to noise in Europe (Engelmann et al., 2026) and Germany [walsch2026].

Additionally, new international research projects and existing exposure assessment tools have expressed interest in adopting healthiar to quantify health impacts.

AI usage disclosure

We only used artificial intelligence (AI) in a few cases for code optimization and debugging, rather than extensively. The whole code was internally reviewed by the two main developers.

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Author contributions

AC and AL are the main authors of healthiar and this manuscript. AC, as creator, led code development and AL was code co-developer.

The rest of the JOSS manuscript authors are contributors to healthiar (methodological advice and systematic testing) and provided feedback on the manuscript. AP, CB and VG are also co-authors of helper functions.

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